EXPERIMENT - 6

Date: 2082/09/16

TO DETERMINE THE REFRACTIVE INDEX OF GLASS SLAB USING TRAVELLING MICROSCOPE.

APPARATUS REQUIRED:

1. Traveling Microscope
2. Glass Slab
3. Lycopodium powder

THEORY:

The refractive index of a medium is defined as

$$\mu = \frac{\text{Real depth (R.D.)}}{\text{Apparent depth (A.D.)}}$$

If O be the real position, O be the apparent position of a mark at the bottom, A be the position of Lycopodium powder on the upper surface of a slab, as focused by the microscope, then

$$\mu = \frac{\text{Real depth (R.D.)}}{\text{Apparent depth (A.D.)}}$$

$$\text{or,} \quad \mu = \frac{3^{rd}\text{ reading} - 1^{st}\text{ reading}}{3^{rd}\text{ reading} - 2^{nd}\text{ reading}}$$

OBSERVATION:

20 division of the main scale = ...1.... cm

1 division of the main scale = ...1/20.... cm

50 Vernier scale divisions coincide with ....49.... main scale divisions.

1 vernier scale division coincide with ....49/50.... main scale divisions.

Vernier constant (V.C.) = (1 m.s.d. - 1 v.s.d.) × value of one m.s.d.

$$= \left(1 - \frac{49}{50}\right) \times \frac{1}{20} \text{ cm} = 0.001 \text{ cm}$$

OBSERVATION TABLE:

No. of Obs.	Reading on the: Bottom O without Slab 1st Reading			Apparent position O' with slab 2nd reading			With the Lycopodium powder on the surface of slab 3rd reading			$\mu = \frac{R.D.}{A.D.} = \frac{x_3 - x_1}{x_3 - x_2}$	Mean μ
	M.S. (cm)	V.S.×V.C. cm	Total x_1 (cm)	M.S. (cm)	V.S.×V.C. cm	Total x_2 (cm)	M.S. (cm)	V.S.×V.C. cm	Total x_3 (cm)		
1	6.15	0.042	6.192	6.75	0.012	6.762	7.95	0.025	7.975	1.469	1.48
2	7.15	0.043	7.193	7.75	0.013	7.763	8.95	0.026	8.976	1.513	
3	8.15	0.044	8.194	8.75	0.014	8.764	9.95	0.027	9.977	1.469	

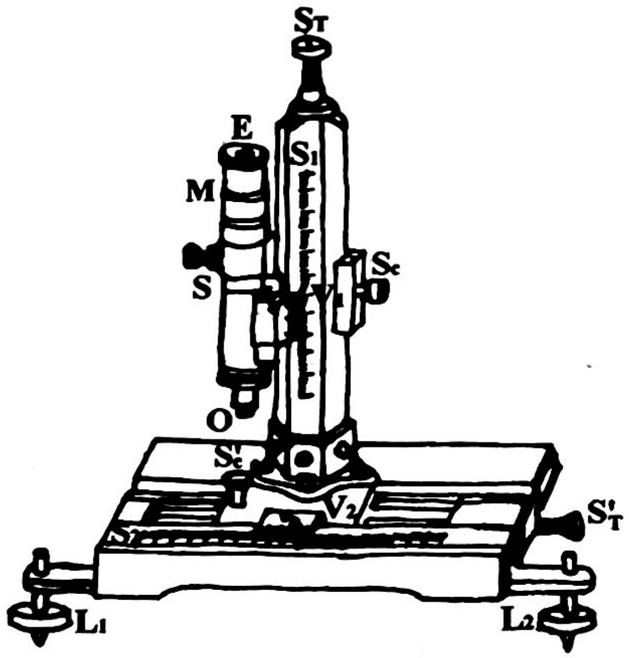

Fig.: Travelling Microscope

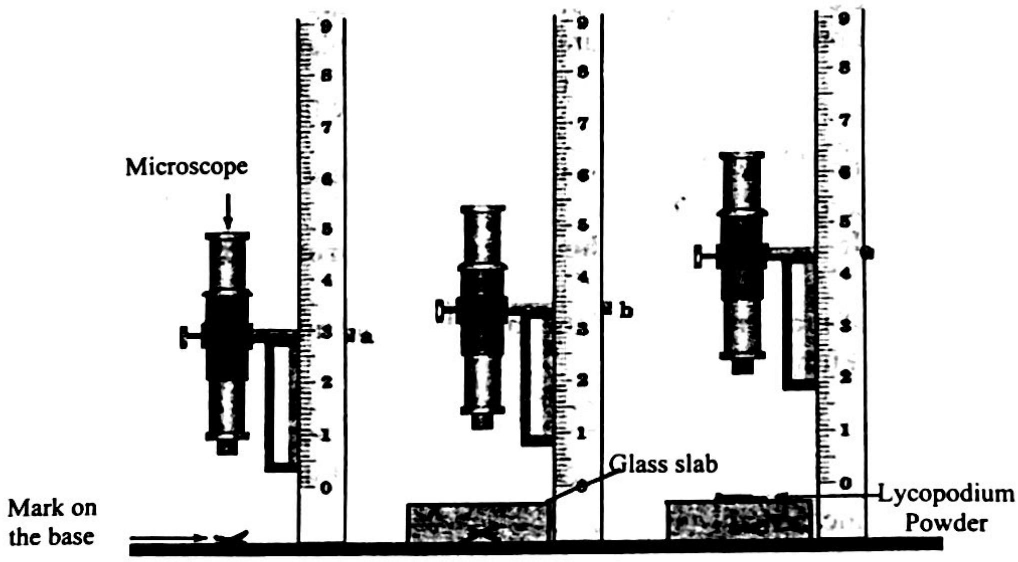

Fig.: Travelling Microscope used to determine refractive index of a glass slab

RESULT:

The value measured for μ is 1.48. Which was found experimentally

PERCENTAGE ERROR:

For 1.4,

Standard value of refractive index (S.V.) = 1.5

$$\% \text{ Error} = \left|\frac{S.V. - O.V.}{S.V.}\right| \times 100\% = \left|\frac{1.5 - 1.4}{1.5}\right| \times 100\%$$

$$= 6.67\%$$

CONCLUSION:

We can find the value of refractive index by using travelling microscope.

SOURCES OF ERROR:

1. The focusing screw may be used after taking the reading with cross marks.
2. The observations may be taken without focusing cross mark correctly.
3. The Vernier scale reading may not be taken carefully.

PRECAUTIONS:

1. The eye should be kept perpendicular to scale while taking HSR and VSR.
2. Focusing screw should be fixed after taking reading with cross marks.
3. Layer of lycopodium powder should be thin.
4. Magnifying glass should be used to take Vernier reading.

VIVA QUESTIONS:

1. What do you mean by refraction of light?
2. What are the laws of refraction?
3. Why is the instrument known as travelling microscope?
4. What are the factors affecting the refractive index of material?
5. What do you understand by absolute and relative refractive index of a medium?
6. Define critical angle?
7. Why is lycopodium powder sprinkled on glass surface? Can you use any other powder?

① Refraction is the phenomenon where a ray of light bends as it passes from one medium to another of different optical density.

② The laws of refraction of light are:

> The incident ray, the refracted ray, and the normal all lie in the same plane at the point of incidence.

> The ratio of sine of incidence to sine angle of refraction is always constant known as refractive index.

③ It is used to measure the dimensions i.e. lengths and heights in upward and downward as well as left and right by moving vertical and horizontal scale known as travelling microscope.

④ The factors affecting the refractive index of material are:

- Nature of the medium
- The colour or wavelength of the incident light

⑤ The ratio of refractive index of the medium to Vacuum is absolute refractive index.

The ratio of refractive index of a medium to another medium is known as relative refractive index.

⑥ The angle of incidence in denser medium for which the angle of refraction in rarer medium is 90° is called critical angle.

⑦ In order to focus the microscope accurately due to the transparency of glass slab, we have used lycopodium powder. Any powder having crystal structure in powder type can be used.

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